

Genotype-dependent zinc partitioning and biochemical responses in tomato under excess zinc

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Highlights

- Zn strongly altered tomato biochemical traits without major growth effects.
- Saint-Pierre accumulated more Zn in leaves than Heinz under excess Zn.
- Heinz retained more Zn in roots at the highest Zn treatment.
- Saint-Pierre showed higher constitutive guaiacol peroxidase activity.
- Tissue Zn was strongly associated with chlorophyll loss across treatments.

Abstract

Zinc (Zn) is an essential micronutrient that becomes phytotoxic when present in excess. This study compared Zn accumulation and biochemical responses in two contrasting tomato genotypes, Heinz and Saint-Pierre (ST-P), exposed to 0, 100 and 250 mg ZnSO₄·7H₂O kg⁻¹ substrate under greenhouse conditions for 55 days. The experiment followed a randomized complete block design with two genotypes, three Zn treatments and three blocks, with three plants per experimental unit. Growth traits, soluble proteins, chlorophylls, carotenoids, tissue Zn, and guaiacol peroxidase (POD) activity were measured. Statistical inference was based on experimental-unit means; the three plants within each experimental unit were treated as subsamples. Zn treatment strongly affected leaf and root soluble proteins, chlorophyll a and b, total chlorophyll, tissue Zn and POD activity, whereas most morphological traits were not significantly affected. Leaf Zn increased to 252.4 ± 2.8 mg kg⁻¹ DW in Heinz and 310.7 ± 12.8 mg kg⁻¹ DW in ST-P at Zn250. Root Zn reached 339.1 ± 44.1 mg kg⁻¹ DW in Heinz and 249.1 ± 18.0 mg kg⁻¹ DW in ST-P, yielding translocation factors of 0.74 and 1.25, respectively. POD activity was constitutively higher in ST-P and increased with Zn in both genotypes. Across nine block × treatment means obtained after averaging the two genotypes, leaf and root Zn were strongly negatively associated with total chlorophyll ($r = -0.967$ and -0.983 , respectively), while root Zn was positively associated with POD activity ($r = 0.984$). These findings indicate genotype-dependent Zn partitioning and antioxidant responses, but the measured variables do not by themselves establish the underlying transporter or subcellular sequestration mechanisms.

Keywords

Solanum lycopersicum; zinc excess; zinc partitioning; translocation factor; peroxidase; chlorophyll; oxidative stress

1. Introduction

Zinc (Zn) is an essential micronutrient required for numerous enzymatic and regulatory functions in plants, but excessive Zn availability can disrupt mineral homeostasis and cellular metabolism. Elevated Zn in agricultural soils can arise from mining, industrial emissions, fertilizers and other anthropogenic inputs. Tomato (*Solanum lycopersicum* L.) is an economically important crop whose genetic diversity can produce substantial differences in responses to abiotic stress.

Excess Zn can affect root development, photosynthetic pigments, protein metabolism and cellular redox balance. Plants respond through a combination of uptake regulation, cell-wall binding, intracellular chelation, compartmentation and antioxidant defenses. Transport systems involving heavy-metal ATPases and metal tolerance proteins contribute to Zn homeostasis, but their relative importance is genotype- and tissue-dependent. Because the present study measured tissue Zn concentrations rather than transporter expression or subcellular localization, these mechanisms are considered as possible explanations rather than demonstrated processes.

The translocation factor (TF), calculated here as the ratio of leaf to root Zn concentration, provides a simple descriptor of Zn partitioning between below- and above-ground tissues. A TF below one indicates

greater Zn retention in roots than leaves, whereas a TF above one indicates greater concentration in leaves. Antioxidant enzymes, including guaiacol peroxidase (POD), can provide complementary information on physiological responses to metal exposure.

Low and moderate metal exposures can sometimes produce non-linear responses, including transient stimulation of selected metabolic processes before inhibition at higher exposure. Such biphasic responses should be interpreted cautiously and should not be equated automatically with a mechanistically demonstrated hormetic process.

This study compared the industrial tomato genotype Heinz and the traditional genotype Saint-Pierre (ST-P) under three $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$ concentrations. We hypothesized that the genotypes would differ in Zn partitioning and antioxidant responses, and that biochemical variables would respond more strongly than gross morphological traits. The objectives were to quantify effects of Zn on growth and water status, characterize changes in soluble proteins and photosynthetic pigments, compare Zn accumulation and partitioning between roots and leaves, and determine whether POD activity was associated with Zn exposure.

2. Materials and methods

2.1. Plant material and experimental conditions

Two tomato genotypes were used: Heinz (Hnz), an industrial determinate-growth cultivar, and Saint-Pierre (ST-P), a traditional indeterminate-growth cultivar. Seeds were surface-sterilized in 5% sodium hypochlorite for 10 min, rinsed three times in sterile distilled water, and germinated in a sterilized peat/sand mixture (1:1, v/v). Seedlings at the 3–5 true-leaf stage (21 days after germination) were transplanted into 1-kg pots filled with substrate and grown under controlled greenhouse conditions with a natural photoperiod and a reported night/day temperature range of 15–20 °C.

2.2. Experimental design and Zn treatments

A randomized complete block design was used with genotype (Heinz and ST-P) and Zn treatment (Zn_0 , Zn_{100} and Zn_{250}) as experimental factors and three blocks. $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$ (purity $\geq 99\%$) was applied at nominal concentrations of 0, 100 and 250 mg kg^{-1} substrate, corresponding to 0, 439 and 1099.5 $\text{mg ZnSO}_4 \cdot 7\text{H}_2\text{O}$ per pot, respectively, via fertigation over four weeks. Each genotype $\times$ treatment combination was represented in each of three independent blocks. Each experimental unit contained three plants, giving 18 experimental units and 54 plants in total. The treatment period was 55 days after transplanting, corresponding to the full-flowering stage.

2.3. Morphological measurements

At harvest, stem length, root length, numbers of ramifications, branches and flowers, fresh weights of roots and shoots, dry weights, and relative water content (RWC) were recorded. Dry weight was determined after oven-drying according to the laboratory protocol. RWC was calculated as $\text{RWC} = [(\text{FW} - \text{DW})/(\text{TW} - \text{DW})] \times 100$, where FW is fresh weight, TW is turgid weight after 24 h in distilled water, and DW is dry weight.

2.4. Biochemical analyses

Leaf and root tissues were sampled at the end of the experiment, immediately placed on ice and processed under refrigerated conditions (4 °C). Soluble proteins were quantified by the Bradford method using bovine serum albumin as the standard, with absorbance measured at 595 nm. Chlorophylls and carotenoids were extracted in 80% acetone (v/v) following Lichtenthaler and Wellburn and quantified spectrophotometrically.

Zn concentrations in leaf and root tissues were determined by flame atomic absorption spectrometry (FAAS) at 213.9 nm after acid digestion with HNO_3 (65%) and H_2O_2 (30%) at 120 °C, using an AAnalyst 400 spectrometer. Quality assurance included procedural blanks, a Zn calibration range of 0.05–5.0 mg L^{-1}

($R^2 = 0.9997$), spike recovery of $97.3 \pm 3.1\%$ ($n = 6$), LOD of 0.02 mg kg^{-1} and LOQ of 0.07 mg kg^{-1} , and analysis of NIST SRM 1573a tomato leaves (measured $\text{Zn} = 30.8 \pm 1.2 \text{ mg kg}^{-1}$ versus certified $30.9 \pm 0.7 \text{ mg kg}^{-1}$). The translocation factor was calculated as $\text{TF} = [\text{Zn}]_{\text{leaf}}/[\text{Zn}]_{\text{root}}$.

2.5. Peroxidase activity

Guaiacol peroxidase (POD; EC 1.11.1.7) activity was measured spectrophotometrically following Chance and Maehly. The 3-mL reaction mixture contained phosphate buffer (0.1 M, pH 7.0), 1% guaiacol, 0.3% H_2O_2 and crude enzyme extract. Absorbance at 470 nm was monitored for 120 s, and the initial linear phase was used for activity calculations. The molar extinction coefficient of tetraguaiacol ($\epsilon = 26.6 \text{ mM}^{-1} \text{ cm}^{-1}$) was used. One unit of activity was defined as the amount of enzyme catalyzing oxidation of $1 \text{ } \mu\text{mol}$ guaiacol min^{-1} under the assay conditions.

2.6. Statistical analysis

Statistical analyses were performed using SPSS v. 26.0 and R v. 4.3.1. Because three plants were nested within each experimental unit, individual plants were treated as subsamples rather than independent biological replicates. For inferential analyses, values from the three plants within each experimental unit were averaged, producing 18 experimental-unit means ($2 \text{ genotypes} \times 3 \text{ treatments} \times 3 \text{ blocks}$). Responses were analyzed using a randomized complete block ANOVA with genotype, Zn treatment and genotype $\times$ treatment interaction as fixed effects and block as a blocking factor. Tukey-adjusted comparisons were used for post hoc mean separation at $\alpha = 0.05$. Partial eta-squared ($\eta^2\text{p}$) was used as an effect-size measure.

Pearson correlations were calculated from experimental-unit means. For pooled correlations, Heinz and ST-P values were first averaged within each block $\times$ treatment combination, yielding nine observations ($3 \text{ blocks} \times 3 \text{ treatments}$; $n = 9$). For genotype-specific correlations, the three experimental units per treatment were retained across the three blocks, giving $n = 9$ per genotype. Thus, no individual plant within an experimental unit was treated as an independent replicate in the correlation analyses.

3. Results

3.1. Morphological responses

Zn exposure did not significantly affect stem length, root length, numbers of ramifications, branches or flowers, or fresh weights in the randomized complete block analysis. Relative water content was also unaffected by genotype, treatment or their interaction (genotype $F = 0.31$, $p = 0.588$; treatment $F = 0.12$, $p = 0.886$; interaction $F = 1.62$, $p = 0.245$). Dry weight showed no significant genotype effect after block adjustment ($F = 4.26$, $p = 0.066$), and no significant treatment or genotype $\times$ treatment effects were detected. Thus, the corrected RCBD analysis supports the conclusion that gross morphology was comparatively stable over the tested exposure range.

3.2. Soluble proteins

Leaf soluble protein concentration was strongly affected by genotype ($F = 675.77$, $p < 0.001$, $\eta^2\text{p} = 0.985$) and Zn treatment ($F = 974.37$, $p < 0.001$, $\eta^2\text{p} = 0.995$), whereas the genotype $\times$ treatment interaction was not significant ($F = 0.81$, $p = 0.470$). Across genotypes, leaf protein increased from $20.83 \pm 3.26 \text{ mg mL}^{-1}$ at Zn0 to $27.91 \pm 3.51 \text{ mg mL}^{-1}$ at Zn100 (+34.0%), then decreased to $14.52 \pm 3.66 \text{ mg mL}^{-1}$ at Zn250 (−30.3% relative to Zn0). Root soluble protein was also strongly affected by genotype ($F = 352.20$, $p < 0.001$, $\eta^2\text{p} = 0.972$) and treatment ($F = 320.44$, $p < 0.001$, $\eta^2\text{p} = 0.985$), with a significant genotype $\times$ treatment interaction ($F = 20.88$, $p < 0.001$, $\eta^2\text{p} = 0.807$). Root protein increased by approximately 62.9% at Zn100 and remained above the control at Zn250 when genotypes were considered together. The biphasic pattern is consistent with a dose-dependent metabolic response, but the present measurements do not identify the specific proteins responsible.

3.3. Zn accumulation and partitioning

Leaf Zn was strongly affected by genotype ($F = 25.90$, $p < 0.001$, $\eta^2p = 0.721$), Zn treatment ($F = 733.93$, $p < 0.001$, $\eta^2p = 0.993$), and their interaction ($F = 10.35$, $p = 0.004$, $\eta^2p = 0.674$). At Zn250, leaf Zn reached $252.4 \pm 2.8 \text{ mg kg}^{-1} \text{ DW}$ in Heinz and $310.7 \pm 12.8 \text{ mg kg}^{-1} \text{ DW}$ in ST-P. Root Zn was likewise affected by genotype ($F = 11.75$, $p = 0.006$, $\eta^2p = 0.540$), treatment ($F = 204.98$, $p < 0.001$, $\eta^2p = 0.976$), and their interaction ($F = 6.83$, $p = 0.014$, $\eta^2p = 0.577$). At Zn250, root Zn was $339.1 \pm 44.1 \text{ mg kg}^{-1} \text{ DW}$ in Heinz compared with $249.1 \pm 18.0 \text{ mg kg}^{-1} \text{ DW}$ in ST-P.

At Zn250, the calculated TF was 0.74 for Heinz and 1.25 for ST-P. This indicates greater root retention in Heinz and greater relative partitioning of Zn to leaves in ST-P. These TF values describe tissue-level partitioning and do not by themselves demonstrate xylem loading, vacuolar sequestration, or transporter activity.

3.4. Photosynthetic pigments

Chlorophyll a was strongly affected by Zn treatment ($F = 82.60$, $p < 0.001$, $\eta^2p = 0.943$), whereas genotype ($F = 1.42$, $p = 0.261$) and genotype $\times$ treatment interaction ($F = 0.24$, $p = 0.794$) were not significant. Chlorophyll a decreased from $19.13 \pm 2.92 \text{ } \mu\text{g mL}^{-1}$ at Zn0 to $13.67 \pm 1.65 \text{ } \mu\text{g mL}^{-1}$ at Zn100 and $9.65 \pm 1.06 \text{ } \mu\text{g mL}^{-1}$ at Zn250, corresponding to reductions of 28.6% and 49.6%, respectively.

Chlorophyll b was also strongly affected by treatment ($F = 17.13$, $p < 0.001$, $\eta^2p = 0.774$), with no significant genotype or interaction effect. Mean chlorophyll b declined from $8.36 \pm 2.76 \text{ } \mu\text{g mL}^{-1}$ at Zn0 to $5.10 \pm 2.18 \text{ } \mu\text{g mL}^{-1}$ at Zn100 and $3.11 \pm 0.86 \text{ } \mu\text{g mL}^{-1}$ at Zn250. Carotenoids were not significantly affected by genotype, treatment or their interaction (treatment $F = 1.49$, $p = 0.272$). Total chlorophyll was strongly affected by Zn treatment ($F = 147.75$, $p < 0.001$, $\eta^2p = 0.967$), without significant genotype or interaction effects.

3.5. Peroxidase activity

POD activity was strongly affected by genotype ($F = 1512.89$, $p < 0.001$, $\eta^2p = 0.993$), Zn treatment ($F = 941.43$, $p < 0.001$, $\eta^2p = 0.995$), and genotype $\times$ treatment interaction ($F = 11.16$, $p = 0.003$, $\eta^2p = 0.691$). Mean POD activity increased in Heinz from 0.077 U mL^{-1} at Zn0 to 0.101 U mL^{-1} at Zn100 and 0.116 U mL^{-1} at Zn250. In ST-P, corresponding values were 0.105, 0.132 and 0.153 U mL^{-1} . Thus, ST-P had approximately 35.5% higher constitutive POD activity at Zn0. Specific POD activity was also significantly affected by genotype, treatment and their interaction ($F = 35.38$, 1365.39 and 48.20, respectively; all $p \leq 0.001$).

3.6. Correlation analysis

Correlation analyses were performed using nine pooled block $\times$ treatment means after averaging Heinz and ST-P within each block $\times$ treatment combination. Leaf Zn was strongly and negatively associated with total chlorophyll ($r = -0.967$, $p < 0.001$, $n = 9$), as was root Zn ($r = -0.983$, $p < 0.001$, $n = 9$). Root Zn was strongly positively associated with POD activity ($r = 0.984$, $p < 0.001$, $n = 9$). In contrast, leaf protein and POD activity were not significantly correlated under the corrected analysis ($r = -0.384$, $p = 0.308$, $n = 9$).

Genotype-specific analyses also showed strong negative associations between leaf Zn and total chlorophyll: Heinz, $r = -0.964$ ($p < 0.001$, $n = 9$); ST-P, $r = -0.948$ ($p < 0.001$, $n = 9$). Because Zn concentration and chlorophyll both changed strongly with treatment, these correlations should be interpreted as associations across the experimental gradient rather than evidence of a direct causal relationship.

4. Discussion

4.1. Biochemical sensitivity despite limited morphological effects

The corrected RCBD analysis indicates that the tested Zn concentrations produced much stronger biochemical than morphological responses. Most growth variables and RWC remained statistically stable, whereas soluble proteins, chlorophylls, tissue Zn and POD activity showed large treatment effects. This dissociation is consistent with the concept that metabolic and biochemical responses can precede conspicuous morphological injury, although the present experiment does not establish a temporal sequence because measurements were made at harvest.

The absence of a significant dry-weight treatment effect after accounting for block is particularly important. The previously reported genotype effect for dry weight ($F = 5.114$, $p = 0.043$) resulted from an analysis that did not account for the blocking structure. With block included in the RCBD model, the genotype effect was not statistically significant ($F = 4.26$, $p = 0.066$). Accordingly, the manuscript no longer interprets dry weight as demonstrating a genotype difference.

4.2. Biphasic protein response

Soluble proteins increased at Zn100 and declined at Zn250, particularly in leaves, while root proteins remained elevated at the highest dose. This pattern is compatible with a biphasic dose response and may reflect transient metabolic adjustment at moderate exposure followed by stronger perturbation at higher exposure. However, because individual stress proteins, protein synthesis rates and proteolysis were not measured, explanations involving metallothioneins, heat-shock proteins or oxidative proteolysis remain hypotheses rather than demonstrated mechanisms.

4.3. Genotype-dependent Zn partitioning

The clearest genotype-dependent response concerned Zn partitioning. Heinz accumulated more Zn in roots at Zn250, whereas ST-P accumulated more Zn in leaves and showed a TF above one. Root retention can reduce delivery of Zn to aerial tissues, whereas greater shoot partitioning may allow different downstream detoxification processes. Cell-wall binding, chelation, xylem loading and vacuolar storage are all plausible mechanisms described for plant Zn homeostasis, but none was directly measured here. Therefore, the present data support a difference in tissue partitioning, not a definitive assignment of transporter-mediated xylem loading or vacuolar sequestration.

The contrasting TF values should also be considered in relation to crop use. A genotype that partitions more Zn to shoots may tolerate soil Zn without major growth effects while simultaneously increasing the potential for Zn entry into edible tissues. Consequently, tolerance and food safety are not necessarily equivalent traits.

4.4. Chlorophyll loss and POD response

Chlorophyll a and b declined substantially with increasing Zn, while genotype effects were not significant. The strong negative associations between tissue Zn and total chlorophyll across the nine pooled means reinforce the link between Zn exposure and pigment loss, but they do not distinguish among possible mechanisms such as altered mineral nutrition, pigment biosynthesis, chloroplast damage or oxidative effects.

POD activity showed a different genotype pattern. ST-P had higher constitutive activity and maintained a higher absolute POD activity at all Zn treatments, while both genotypes increased POD with Zn exposure. The strong positive association between root Zn and POD across the pooled treatment gradient is compatible with induction of antioxidant capacity as Zn accumulation increases. Nevertheless, ROS concentrations and additional antioxidant enzymes were not measured, so the data demonstrate POD responsiveness rather than a complete characterization of oxidative stress.

Importantly, the corrected correlation analysis no longer supports a significant relationship between leaf soluble protein and POD activity. The previous association was therefore removed from the manuscript.

This illustrates the importance of basing correlation inference on independent experimental units rather than pooled plant-level observations.

4.5. Implications for genotype selection

Taken together, the results identify two measurable phenotypic dimensions of Zn response: tissue partitioning and antioxidant capacity. ST-P combined greater shoot Zn partitioning with higher constitutive POD activity, whereas Heinz retained more Zn in roots at the highest treatment. These traits may be useful as candidate indicators in comparative screening, but their predictive value for field tolerance requires validation across additional genotypes, Zn regimes and environmental conditions.

Future experiments should directly quantify Zn localization at the cellular and subcellular levels and examine expression or activity of candidate transport systems such as HMA and MTP proteins. Such measurements would test whether the observed TF contrast is associated with differential xylem loading, vacuolar sequestration or other processes. Measurements of ROS, membrane damage and a broader antioxidant panel would also help determine how Zn partitioning is connected to redox regulation.

5. Conclusions

Excess Zn produced pronounced biochemical responses in Heinz and ST-P tomato despite limited effects on gross morphology and water status. Zn strongly altered soluble proteins, chlorophylls, tissue Zn accumulation and POD activity. At Zn250, Heinz retained more Zn in roots (TF = 0.74), whereas ST-P showed greater relative partitioning to leaves (TF = 1.25) and higher constitutive POD activity. The corrected experimental-unit analysis further showed strong negative associations between tissue Zn and total chlorophyll and a strong positive association between root Zn and POD activity across the experimental gradient. These results support genotype-dependent differences in Zn partitioning and antioxidant responsiveness, while avoiding attribution of unmeasured transporter or subcellular mechanisms. Further physiological and molecular studies are needed to determine how these phenotypes arise and whether they can be exploited reliably in tomato breeding.

Tables

Variable	Genotype F (p)	η^2p	Zn treatment F (p)	η^2p	G × T F (p)	η^2p
Dry weight	4.26 (0.066)	0.299	0.48 (0.630)	0.088	0.38 (0.692)	0.071
RWC	0.31 (0.588)	0.030	0.12 (0.885)	0.024	1.62 (0.245)	0.245
Leaf soluble protein	675.77 (<0.001)	0.985	974.37 (<0.001)	0.995	0.81 (0.470)	0.140
Root soluble protein	352.20 (<0.001)	0.972	320.44 (<0.001)	0.985	20.88 (<0.001)	0.807
Leaf Zn	25.90 (<0.001)	0.721	733.93 (<0.001)	0.993	10.35 (0.004)	0.674
Root Zn	11.75 (0.006)	0.540	204.98 (<0.001)	0.976	6.83 (0.014)	0.577
Chlorophyll a	1.42 (0.261)	0.125	82.60 (<0.001)	0.943	0.24 (0.794)	0.045
Chlorophyll b	0.38 (0.550)	0.037	17.13 (<0.001)	0.774	0.30 (0.749)	0.056
Total chlorophyll	0.14 (0.716)	0.014	147.75 (<0.001)	0.967	0.50 (0.620)	0.091
Carotenoids	0.41 (0.536)	0.039	1.49 (0.272)	0.229	0.35 (0.716)	0.065
POD activity	1512.89 (<0.001)	0.993	941.43 (<0.001)	0.995	11.16 (0.003)	0.691
POD specific activity	35.38 (<0.001)	0.780	1365.39 (<0.001)	0.996	48.20 (<0.001)	0.906

Table 1. Corrected randomized complete block ANOVA based on 18 experimental-unit means (three plants averaged within each experimental unit). Block was included as a blocking factor. η^2p , partial eta-squared.

Variable	Genotype	Treatment	Mean ± SD	Unit
Leaf soluble protein	Heinz	Zn0	17.831 ± 0.643	mg mL ⁻¹
Leaf soluble protein	Heinz	Zn100	24.618 ± 0.800	mg mL ⁻¹
Leaf soluble protein	Heinz	Zn250	11.144 ± 0.411	mg mL ⁻¹
Leaf soluble protein	ST-P	Zn0	23.836 ± 0.277	mg mL ⁻¹
Leaf soluble protein	ST-P	Zn100	31.192 ± 0.485	mg mL ⁻¹
Leaf soluble protein	ST-P	Zn250	17.888 ± 0.315	mg mL ⁻¹
Root soluble protein	Heinz	Zn0	22.847 ± 0.629	mg mL ⁻¹
Root soluble protein	Heinz	Zn100	37.649 ± 1.875	mg mL ⁻¹
Root soluble protein	Heinz	Zn250	24.858 ± 0.432	mg mL ⁻¹
Root soluble protein	ST-P	Zn0	29.037 ± 0.271	mg mL ⁻¹
Root soluble protein	ST-P	Zn100	46.897 ± 1.371	mg mL ⁻¹
Root soluble protein	ST-P	Zn250	39.369 ± 0.806	mg mL ⁻¹
Leaf Zn	Heinz	Zn0	25.851 ± 5.145	mg kg ⁻¹ DW
Leaf Zn	Heinz	Zn100	214.176 ± 14.075	mg kg ⁻¹ DW
Leaf Zn	Heinz	Zn250	252.400 ± 2.782	mg kg ⁻¹ DW
Leaf Zn	ST-P	Zn0	20.066 ± 0.788	mg kg ⁻¹ DW
Leaf Zn	ST-P	Zn100	251.053 ± 20.292	mg kg ⁻¹ DW
Leaf Zn	ST-P	Zn250	310.698 ± 12.787	mg kg ⁻¹ DW
Root Zn	Heinz	Zn0	25.604 ± 4.618	mg kg ⁻¹ DW
Root Zn	Heinz	Zn100	226.252 ± 23.272	mg kg ⁻¹ DW
Root Zn	Heinz	Zn250	339.089 ± 44.085	mg kg ⁻¹ DW
Root Zn	ST-P	Zn0	33.577 ± 6.948	mg kg ⁻¹ DW
Root Zn	ST-P	Zn100	196.080 ± 1.927	mg kg ⁻¹ DW
Root Zn	ST-P	Zn250	249.108 ± 17.964	mg kg ⁻¹ DW
Chlorophyll a	Heinz	Zn0	18.853 ± 0.951	µg mL ⁻¹
Chlorophyll a	Heinz	Zn100	13.024 ± 0.797	µg mL ⁻¹
Chlorophyll a	Heinz	Zn250	9.494 ± 0.794	µg mL ⁻¹
Chlorophyll a	ST-P	Zn0	19.413 ± 2.526	µg mL ⁻¹
Chlorophyll a	ST-P	Zn100	14.316 ± 0.959	µg mL ⁻¹
Chlorophyll a	ST-P	Zn250	9.807 ± 0.809	µg mL ⁻¹
Chlorophyll b	Heinz	Zn0	8.982 ± 0.886	µg mL ⁻¹
Chlorophyll b	Heinz	Zn100	5.230 ± 0.500	µg mL ⁻¹
Chlorophyll b	Heinz	Zn250	3.050 ± 0.809	µg mL ⁻¹
Chlorophyll b	ST-P	Zn0	7.748 ± 2.463	µg mL ⁻¹
Chlorophyll b	ST-P	Zn100	4.965 ± 2.340	µg mL ⁻¹
Chlorophyll b	ST-P	Zn250	3.174 ± 0.325	µg mL ⁻¹
Total chlorophyll	Heinz	Zn0	27.835 ± 1.619	µg mL ⁻¹
Total chlorophyll	Heinz	Zn100	18.254 ± 0.302	µg mL ⁻¹
Total chlorophyll	Heinz	Zn250	12.544 ± 0.754	µg mL ⁻¹
Total chlorophyll	ST-P	Zn0	27.160 ± 2.201	µg mL ⁻¹
Total chlorophyll	ST-P	Zn100	19.281 ± 1.382	µg mL ⁻¹
Total chlorophyll	ST-P	Zn250	12.980 ± 1.121	µg mL ⁻¹
POD activity	Heinz	Zn0	0.077 ± 0.001	U mL ⁻¹
POD activity	Heinz	Zn100	0.101 ± 0.002	U mL ⁻¹
POD activity	Heinz	Zn250	0.116 ± 0.003	U mL ⁻¹
POD activity	ST-P	Zn0	0.105 ± 0.001	U mL ⁻¹
POD activity	ST-P	Zn100	0.132 ± 0.001	U mL ⁻¹

Variable	Genotype	Treatment	Mean $\pm$ SD	Unit
POD activity	ST-P	Zn250	0.153 $\pm$ 0.002	U mL ⁻¹

Table 2. Means $\pm$ SD calculated from the three independent experimental units within each genotype $\times$ treatment combination. For biochemical variables, each experimental-unit value is the mean of three subsampled plants.

Variables	Pearson r	p	n	Analysis
Leaf Zn $\times$ total chlorophyll	-0.967	<0.001	9	Pooled block $\times$ treatment means
Root Zn $\times$ total chlorophyll	-0.983	<0.001	9	Pooled block $\times$ treatment means
Root Zn $\times$ POD	+0.984	<0.001	9	Pooled block $\times$ treatment means
Leaf protein $\times$ POD	-0.384	0.308	9	Pooled block $\times$ treatment means
Heinz: leaf Zn $\times$ total chlorophyll	-0.964	<0.001	9	Genotype-specific
ST-P: leaf Zn $\times$ total chlorophyll	-0.948	<0.001	9	Genotype-specific

Table 3. Corrected Pearson correlations based on experimental-unit means. For pooled analyses, Heinz and ST-P values were averaged within each block $\times$ treatment combination.

Supplementary-style calculated descriptors

Genotype	Zn treatment	Leaf Zn (mg kg ⁻¹ DW)	Root Zn (mg kg ⁻¹ DW)	TF
Heinz	Zn250	252.4	339.1	0.74
ST-P	Zn250	310.7	249.1	1.25

TF = leaf Zn/root Zn. These values are descriptive partitioning indices and do not establish a specific cellular mechanism.

Figure legends

Figure 1. Morphological and water-status responses of Heinz and Saint-Pierre tomato plants under Zn0, Zn100 and Zn250 treatments. Values are based on experimental-unit means; error bars should represent SD or SE consistently with the final artwork.

Figure 2. Soluble protein concentrations and Zn accumulation in leaves and roots of Heinz and Saint-Pierre under Zn stress. The inset reports the translocation factor (TF = leaf Zn/root Zn).

Figure 3. Effects of Zn treatment on chlorophyll a, chlorophyll b, carotenoids and guaiacol peroxidase activity in Heinz and Saint-Pierre.

Figure 4. Relationships among tissue Zn, total chlorophyll and POD activity based on corrected experimental-unit means. Pooled correlations use nine block $\times$ treatment means.

Statements and declarations

Competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Author contributions

A.A. conceptualized the study, conducted the experiments, performed data analysis and wrote the original draft. Y.N. and M.A. supervised the research, provided methodological guidance, reviewed and edited the manuscript, and contributed to resource acquisition. M.A. and A.T. contributed to biochemical assays and data interpretation. All authors read and approved the final manuscript.

Data availability

The datasets generated and analyzed during the current study are available from the corresponding author on reasonable request.

Ethics approval

This article does not contain any studies with human participants or animals performed by any of the authors.

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Author submission note — statistical correction

The manuscript above uses the supplied Stats_Stress_Zn_Tomate_COMPLET.xlsx as the data source and treats the 18 block × genotype × treatment combinations as experimental units. The three plants nested within each experimental unit were averaged before inferential analysis. The pooled correlation analysis uses nine observations because Heinz and ST-P were averaged within each block × treatment combination, exactly as confirmed by the authors.

Before journal submission, the authors should verify the FAAS quality-control values, greenhouse details, funding statement, author contributions and figure artwork against laboratory records and the final submission package. No unverified cultivar lot/accession identifiers are included in this version.